

Zhifei Li

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Research Interests

My research interests lie broadly in **machine learning systems**, spanning efficient infrastructure, training and inference systems, and the design of systems for emerging AI workloads. I am also interested in exploring how AI techniques can advance systems design.

Research Experience

Sky Computing Lab

University of California, Berkeley

RESEARCH INTERN, ADVISED BY [PROF. ION STOICA](#)

WORKED WITH [PROF. JOSEPH E. GONZALEZ](#), [PROF. MATEI ZAHARIA](#)

July 2025 - December 2025

- **SkyNomad: Multi-Region Spot Instance Scheduling** (accepted to NSDI '27)
 - Designed a multi-region spot instance scheduling system, addressing single-region availability bottlenecks for offline workloads, via Unified Cost Model trading off cross-region availability and pricing vs. migration costs
 - Achieved >50% cost reduction over previous SOTA (NSDI '24 best paper), saved \$1,000+ from a \$2,200 training job vs. AWS SageMaker
 - Led project from research formulation to production, drove methodology design, and built simulation framework
- **LEANN: Storage-Efficient Compound AI Systems** (MLSys '26)
 - Co-designed a two-level recompute algorithm to cut vector index storage overhead in RAG pipelines
 - Achieved 97% storage reduction, <5% latency impact; led open-source implementation to [12,000+ GitHub stars](#)
 - Built the system extending FAISS C++, contributing 70% codebase; drove evaluation efforts
- **AI-driven Systems Research**
 - Investigated systems optimization through evolutionary algorithms and LLM-guided design space exploration
 - Led case study in [Barbarians at the Gate](#) paper, which demonstrated 30% improvement over the SOTA
 - Co-developed [FrontierCS](#) benchmark with problem specifications and evaluations for 40 open-ended problems

Education

University of California, Berkeley

Berkeley, CA, USA

EXCHANGE STUDENT, COMPUTER SCIENCE

August 2024 - December 2024

- **CS294-162 Machine Learning Systems** graduate seminar
 - Optimized complex DAG workload execution through intelligent data placement and cross-cloud task scheduling
 - Achieved 45% cost reduction; select optimizations merged into [SkyPilot](#)

Renmin University of China (Ranked 23rd globally on [CSRankings 2025](#))

Beijing, China

BACHELOR'S IN COMPUTER SCIENCE, **TURING HONORS CLASS**

September 2022 - Expected: 06/26

- GPA: 3.8/4.0 (Top 5%)

Publications

SkyNomad: On Using Multi-Region Spot Instances to Minimize AI Batch Job Cost

Zhifei Li*, *Tian Xia**, *Ziming Mao*, *Zihan Zhou*, *Ethan J. Jackson*, *Jamison Kerney*, *Zhanghao Wu*, *Pratik Mishra*, *Yi Xu*, *Yifan Qiao*, *Ion Stoica*
NSDI '27; [\[PDF\]](#)

LEANN: A Low-Storage Overhead Vector Index

Yichuan Wang, **Zhifei Li**, *Shu Liu*, *Yongji Wu*, *et al.*, *Ion Stoica*, *Sewon Min*, *Matei Zaharia*, *Joseph Gonzalez*
MLSYS '26 (BEST PAPER AWARD); [\[PDF\]](#)

Barbarians at the Gate: How AI is Upending Systems Research

*Audrey Cheng**, *Shu Liu**, *Melissa Pan**, **Zhifei Li**, *Bowen Wang*, *Alex Krentsel*, *et al.*, *Matei Zaharia*, *Ion Stoica*
ARXIV PREPRINT; [\[PDF\]](#)

FrontierCS: Evolving Challenges for Evolving Intelligence

Qiuyang Mang*, Wenhao Chai*, Zhifei Li*, Huanzhi Mao*, et al., Ion Stoica, Jingbo Shang, Zhuang Liu, Alvin Cheung
ICML '26; [PDF]; [WEBSITE]

SkyWalker: A Locality-Aware Cross-Region Load Balancer for LLM Inference

Tian Xia, Ziming Mao, Jamison Kerney, Ethan J. Jackson, Zhifei Li, Jiarong Xing, Scott Shenker, Ion Stoica
EUROSYS 2026; [PDF]

Open-Source Projects

LEANN: the Smallest Vector Index in the World

 [LEANN](#) 12k ★

ENJOY 97% STORAGE SAVINGS FOR RAG APPLICATION ON YOUR PERSONAL DEVICE September 2024 - Present

- Led research-to-production translation of [LEANN](#) from prototype to production-ready open-source Python package with CI/CD pipeline, grew to [12,000+ GitHub stars](#) with 20 active external contributors and 40k+ downloads
- Drove technical outreach including blog posts social media campaign achieving 600k+ views

SkyPilot: Run AI on Any Infra

 [SkyPilot](#) 9.1k ★

FRAMEWORK FOR RUNNING ML/AI WORKLOADS ACROSS ANY CLOUD INFRASTRUCTURE September 2024 - Present

- Top 10 contributor; created 70+ issues and merged 50+ pull requests; contributed 30,000+ lines of code changes
- Implemented [High Availability Controller](#) for SkyServe control plane; adopted by startups including Hypermode, IP Copilot and Liner

Services

USENIX ATC '25 Artifact Evaluation Committee

REVIEWER May 2025

Introduction to Computer Systems (ICS)

HEAD TEACHING ASSISTANT Fall 2024, Spring 2025

- Led 6 TAs in teaching systems curriculum that covered cache hierarchies and memory optimization to 200+ students

RUC Computer Association

PRESIDENT July 2024 - July 2025

- Organized 10+ tech talks, 5000+ total attendees, covering topics from Functional Programming to Rust ecosystem
- Led 100+ members across 6 departments, fostered a startup atmosphere and inclusive environment

Cheese Tech

CO-FOUNDER September 2023 - October 2024

- Raised \$300K seed funding, grew to 1,000+ users across 3 partner institutions within first year
- Developed AI-powered research platform with inspiration tracking, progress management, and intelligent advising

Honors and Awards

First Place, National System Capability Competition (among 10,000 from 1,200 schools) November 2025

Elite Collegiate Award, China Computer Federation (<100 recipients nationally) August 2025

Dean's Scholarship, Gaoling School of AI (15 recipients out of 2,000) May 2025

National Scholarship (Top 0.2% nationally) September 2024

First-Class Scholarship for Social Service (48 recipients out of 35,000) September 2023

First Prize, National Olympiad in Informatics, Beijing (300 winners) December 2019

Skills

- Coding** C++, CUDA, Python, Rust, TypeScript
- Tools** PyTorch, SkyPilot, Ray, Kubernetes, verl, NeMo, Nix, Fish, Typst
- Languages** English, Chinese (native), French
- Hobbies** Texas Hold'em, Badminton, Chess, Cooking